

# Peer review and response

Project: Reproducible AI Research Study

## 1. Your mission

### Your learning target

I can explain the mechanism and terminology of peer review and response.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

### Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Writing results and limitations. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Figures that preserve evidence. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Inspecting failures and anomalies. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

This week you will practise:

Explain the mechanism and terminology of peer review and response

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

### Retrieve what you already know

Recall Writing results and limitations. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

### My starting explanation

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### This week's end goal

Explain peer review and response using the records and boundaries of this fictional investigation.

## 2. Notice the evidence

Peer review challenges evidence, method and clarity. Responses should explain changes with artifacts rather than merely defend the original wording.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

**List the given facts and one fact that is not supplied.**

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## 3. See how it works

For each comment record concern, response, change location and new evidence if needed. If no new evidence exists, narrow the claim instead of promising an unperformed test.

### Peer review and response — evidence trace

**Input and scope · Check or calculation · Conclusion and limit**

**Input and scope:** Reviewer asks for a fitted-line baseline; author responds that the neural network is advanced, without a comparison.

**Check or calculation:** For each comment record concern, response, change location and new evidence if needed. If no new evidence exists, narrow the claim instead of promising an unperformed test.

**Conclusion and limit:** The response lacks evidence. Add a fair comparison under the plan or narrow the claim to the existing baseline; record any new analysis as such.

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The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

**Connect the picture and the explanation**

Point to the input, the deciding step and the result in this week’s visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

**4. Compare the cases**

**Compare a familiar case with a changed case**

Reviewer asks for a fitted-line baseline; author responds that the neural network is advanced, without a comparison. Show the intermediate steps and give the conclusion with its scope.

**Keep the procedure fixed. Explain what changed in the inputs and result.**

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**Change one thing in the pictured case**

Change: The team only changes "photos" to "plants" without revisiting analysis.

Prediction: The substantive independence issue remains unresolved.

Reason: Renaming records does not change their shared source objects.

**5. Words that unlock the idea**

**Word · Meaning**

**Word:** Review comment

**Meaning:** A concern about evidence, method or presentation

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**Word:** Response record

**Meaning:** A mapping from comment to change or justified disagreement

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**Word:** Claim narrowing

**Meaning:** Reducing a statement to the evidence actually available

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**Use two terms to explain a step in this week's example.**

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## 6. Follow a worked example

Reviewer asks for a fitted-line baseline; author responds that the neural network is advanced, without a comparison.

The response lacks evidence. Add a fair comparison under the plan or narrow the claim to the existing baseline; record any new analysis as such.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

### Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

### A pictorial worked case: Peer review and response

Respond to a reviewer with evidence. A reviewer asks why four photos of one plant were treated as four independent plants.

### The facts we will use

The original table contains four views of P1 and four views of P2: eight photos from two plants.

## Peer review and response

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



#### 1. Identify the concern

The reviewer questions the unit of analysis.



#### 2. Check the records

Plant identifiers confirm two plants, not eight independent plants.



#### 3. Revise transparently

Correct the description and revise any analysis that assumed independence.

### CHANGE ONE THING

The team only changes "photos" to "plants" without revisiting analysis.

Explain the changed result using the matching lesson.

## Read the picture one step at a time

### Step · What happens and why

**Step:** 1. Identify the concern

**What happens and why:** The reviewer questions the unit of analysis.

**Step:** 2. Check the records

**What happens and why:** Plant identifiers confirm two plants, not eight independent plants.

### Step: 3. Revise transparently

**What happens and why:** Correct the description and revise any analysis that assumed independence.

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Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

Peer review should connect a criticism to the actual records and a concrete correction or reasoned response. Polite language alone cannot repair a methodological error.

### What this example does not establish

Two plants provide limited coverage; grouping correctly does not create more independent evidence.

Read the labelled picture: Count photos and underlying plants in the reviewer's concern. What analysis assumption needs revision, and why does changing only the word photos to plants fail to repair it?

Change one thing: The team only changes "photos" to "plants" without revisiting analysis. Predict the result before checking the explanation. What remains the same?

## 7. Find and repair a mistake

Agreement with a reviewer is not required, but a disagreement needs explicit reasoning and evidence.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself. The first may require a correction; the second requires an explicitly revised method and new evidence. Preserve unsuccessful cases so an improved result cannot hide the original problem.

**Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

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### Locate the earliest unsupported step

A report omits this qualification: Agreement with a reviewer is not required, but a disagreement needs explicit reasoning and evidence. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

**State one repair and the checks you would rerun.**

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## 8. Practise together

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

**Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

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**Explain your prediction to a partner before comparing results.**

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### Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Peer review and response”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

**Plan · Do · Check**

**Plan:** What is given? Which rule or procedure applies?

**Do:** Record each intermediate step.

**Check:** Does the result follow? What remains unknown?

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Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

## Check your reasoning

Use the worked case for Peer review and response. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

## Explain the picture

Count photos and underlying plants in the reviewer's concern. What analysis assumption needs revision, and why does changing only the word photos to plants fail to repair it?

### My explanation using a pictured step and the given facts

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This is supported practice. Keep the separately named independent case for an attempt before feedback.

## 9. Run the investigation

### Predict, investigate and preserve the evidence

Reviewer asks for a fitted-line baseline; author responds that the neural network is advanced, without a comparison. Show the intermediate steps and give the conclusion with its scope.

### Record · Your evidence

**Record:** Given input and fixed rule

**Your evidence:** \_\_\_\_\_

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**Record:** Prediction before checking

**Your evidence:** \_\_\_\_\_

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**Record:** Actual result, or “not run”

**Your evidence:** \_\_\_\_\_

\_\_\_\_\_

**Record:** First mismatch and one explanation

**Your evidence:** \_\_\_\_\_

\_\_\_\_\_

**Record:** Boundary or missing input and expected response

**Your evidence:** \_\_\_\_\_

\_\_\_\_\_

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

### Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

## 10. Build your understanding

### Explain → apply → challenge

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

A report omits this qualification: Agreement with a reviewer is not required, but a disagreement needs explicit reasoning and evidence. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

**Explain which evidence supports your conclusion and which claim remains unproved.**

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## 11. Connect your project

Add a labelled peer review and response evidence sheet to Reproducible AI Research Study. Show how you can explain the mechanism and terminology of peer review and response. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

### Project evidence · Record

**Project evidence:** Component and version

**Record:** \_\_\_\_\_

\_\_\_\_\_

**Project evidence:** Inputs permitted at prediction time

**Record:** \_\_\_\_\_

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**Project evidence:** Trace and comparison with earlier work

**Record:** \_\_\_\_\_

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**Project evidence:** Failure, limit and human review

**Record:** \_\_\_\_\_

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### Evidence for your project

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

## 12. Show what you know

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer. For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside

any revision.

**Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

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For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

**My limited claim and a useful next check**

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## 13. Study a complete case

### Complete evidence record

### Peer review and response — evidence trace

**Input and scope · Check or calculation · Conclusion and limit**

**Input and scope:** Reviewer asks for a fitted-line baseline; author responds that the neural network is advanced, without a comparison.

**Check or calculation:** For each comment record concern, response, change location and new evidence if needed. If no new evidence exists, narrow the claim instead of promising an unperformed test.

**Conclusion and limit:** The response lacks evidence. Add a fair comparison under the plan or narrow the claim to the existing baseline; record any new analysis as such.

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The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for

qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the input through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

### **My complete trace of the worked case**

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## **14. Try a new case independently**

### **Independent case: Apply a new case independently**

Reviewer notes missing failure counts; author deletes the limitation paragraph. What is an adequate response?

### **My independent explanation, trace and limits**

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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

### **Attempt this without hints**

Reviewer notes missing failure counts; author deletes the limitation paragraph. What is an adequate response?

### **My independent procedure, result and limitation**

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Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

**Use feedback to improve**

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

**15. Your lesson route**

Start with this question: Respond to a reviewer with evidence. A reviewer asks why four photos of one plant were treated as four independent plants.

**By the end, I can · How I will show it**

**By the end, I can:** Explain the mechanism and terminology of peer review and response

**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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**By the end, I can:** Apply the stated procedure to a new case with a traceable result

**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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**By the end, I can:** Identify an assumption or limitation and design a revealing follow-up

**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

## 16. Finish and return

Close your notes. Explain this goal in your own words: Explain the mechanism and terminology of peer review and response

### My deciding step, evidence and one remaining question

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If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.

## 17. Supported practice, repair and portfolio

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Respond to a reviewer with evidence. A reviewer asks why four photos of one plant were treated as four independent plants.

### Practice 1: explain the deciding evidence

Which labelled step matches this detail? Correct the description and revise any analysis that assumed independence.

1. Revise transparently
2. Check the records
3. Identify the concern

### My selected conclusion, deciding fact and explanation

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Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Revise transparently Correct the description and revise any analysis that assumed independence. Peer review should connect a criticism to the actual records and a concrete correction or reasoned response. Polite language alone cannot repair a methodological error.

**Practice 2: explain the deciding evidence**

Changed-case practice: The team only changes "photos" to "plants" without revisiting analysis. Which conclusion is supported by the supplied facts?

- 1. This one example guarantees correct results for every future input.
- 2. The substantive independence issue remains unresolved.
- 3. We can decide the result without examining the changed input or the procedure.

**My selected conclusion, deciding fact and explanation**

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Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The substantive independence issue remains unresolved. Renaming records does not change their shared source objects.

**Repair a missing building block**

Review Writing results and limitations (ai-7/31). Goal: Explain the mechanism and terminology of writing results and limitations. Short practice: Compute the supported-answer rates for 9/12 and 10/12. What actual outcome was measured, and why is a claim that every student becomes smarter outside this evidence?. Return to this week and explain what you changed.

Review Figures that preserve evidence (ai-7/29). Goal: Explain the mechanism and terminology of figures that preserve evidence. Short practice: Draw count bars for 6/10 and 8/10 using the stated axis. Why must the denominator and fixed task context remain visible, and what distortion arises from an undisclosed axis

start at five?. Return to this week and explain what you changed.

Review Inspecting failures and anomalies (ai-7/26). Goal: Explain the mechanism and terminology of inspecting failures and anomalies. Short practice: Calculate mean and median for 2,2,2,10. What does the timeout log establish, and why is deleting that inconvenient record incompatible with the stated plan?. Return to this week and explain what you changed.

### **Keep your completed work**

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.